Earth and Planetary Science Letters, 94 (1989) 123–130
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands

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Paleomagnetic results from Late Paleozoic dikes from the northwestern Junggar Block, northwestern China

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Received February 20, 1989; revised version received May 19, 1989

Paleomagnetic data have been obtained from Late Paleozoic mafic and silicic dikes in northwestern Junggar, a triangular fragment of Early to Middle Paleozoic trapped oceanic crust located between the Kazakhstan, Tarim and Siberian Blocks. Discordance between the observed declination and expected declinations from Tarim and Kazakhstan suggests a 30–40° counterclockwise tectonic rotation of the sampling region relative to Tarim and Kazahkstan. This rotation is attributed to post-Triassic thrust and strike-slip faulting. At face value, there is a good agreement between the observed mean inclination and those expected from Kazakhstan and Tarim. Tectonic interpretation of this mean inclination is obscured by a large difference in inclination from two sampling localities. A combination of geomagnetic secular variation and local tilt of fault block may cause this dispersion.

1. Introduction

The Junggar Block of northwestern China covers an area of nearly 130,000 km^2 (Fig. 1). The block appears to overlie a triangular fragment of Early to Middle Paleozoic ocean crust containing volcanic arc remnants and probably small slices of older crystalline basement rocks. This oceanic crust was trapped between the Siberian, Kazakhstan and Tarim Blocks during the consolidation of Asia as they adjusted to their irregular boundaries [1]. Because of its location, the Junggar Block is one of the keys in understanding the complex history of accretion and amalgamation of the blocks that now form central Asia. Although it has generally been accepted that the Junggar Block collided with the Tarim and Kazakhstan Blocks in the Late Paleozoic [1–3], there are no published paleomagnetic data from Junggar to verify this hypothesis. In this paper we present new paleomagnetic data obtained from Late Paleozoic dikes near Keramayi to help constrain the Late Paleozoic paleogeography of the Junggar Block (Fig. 2).

2. Geologic setting

Feng et al. [1] present a detailed review of the regional geology of west Junggar. Vertically dipping mafic and silicic dikes, which we sampled during the summer of 1988, intrude Late Carboniferous granitic plutons in western Junggar. These granitic plutons have Permian K-Ar ages and Late Carboniferous U-Pb ages [4] and intrude Paleozoic rock assemblages in regions both north and south to the Hueshentaolege Valley, a Late Paleozoic suture between Kazakhstan and Junggar [1]. These plutons mark the final stitching of the Kazakhstan and Junggar Blocks as the ocean separating them closed in the Late Paleozoic. The closure of this ocean seems to be documented also in the Upper Paleozoic regressive sequence in southern Junggar, which shows Upper Carboniferous turbidites grading into Lower Permian marine sandstones and Upper Permian fluvial and lacustrine deposits [5]. The dikes do not cut overlying Mesozoic and Cenozoic fluvial and lacustrine sequences. We believe that the age of dikes is very close to that of the Late Carbonif-

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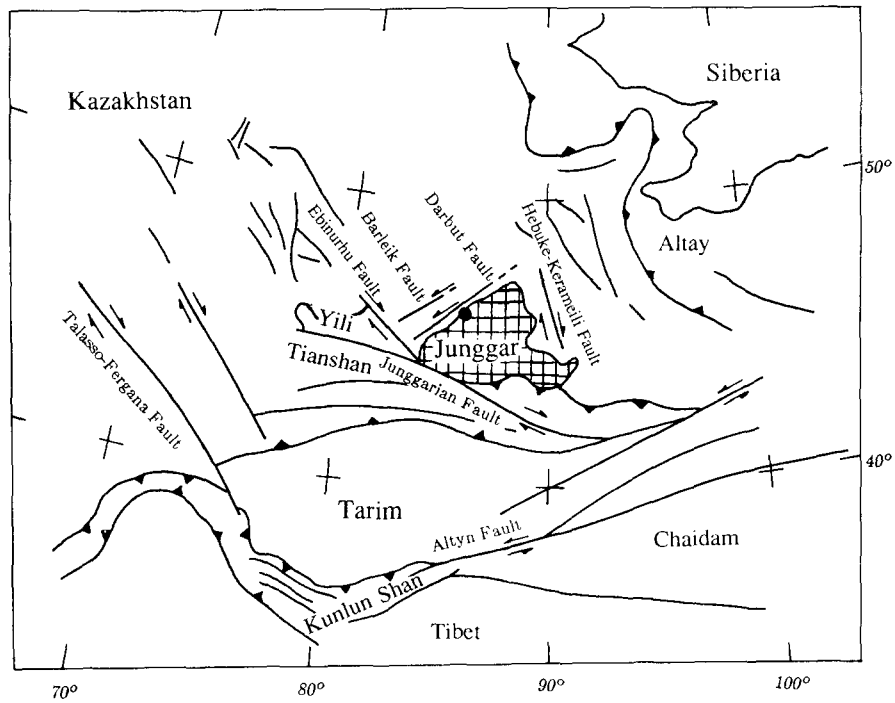


Fig. 1. Generalized tectonic map of northwestern China. Circle indicates the sampling region.

erous granitic plutons, but not later than Permian.

Strike-slip faults developed during the Late Carboniferous and Permian as Siberian, Kazakhstan and Tarim Blocks adjusted to their irregular boundaries [2,3]. Several northeast-trending left-natural faults (e.g. the Darbut, Tuoli and Barleik faults) are close to the sampling region (Fig. 2). Because of an irregular geometrical fit between the Siberian, Tarim and Kazakhstan Blocks, the Junggar Block was probably protected from substantial compressional shortening and imbrication during the consolidation of central Asia. However, the original configuration of west Junggar has been complicated by reactivation of the strike-slip faults and development of new conjugate faults within the Tianshan fold belt during the Cenozoic Himalayan orogeny [6].

3. Paleomagnetism

Using portable drilling equipment, we collected oriented samples from Late Paleozoic dike swarms at three localities. Samples were oriented by a magnetic and sun compass. Two diabase dike swarms are located about 20 km northwest of Keramayi (Fig. 2), and intruded the same pluton. The distance between the first and second swarms is about 3 km. The third dike swarm is located in a small granitic body of less than 30 km^2, 50 km southwest of Keramayi. The majority of the dikes at this location are diabase, but a few are silicic. We collected six cores from each of 25 dikes, each defined as an individual cooling unit (site).

The majority of the specimens were treated by step-wise alternating field (AF) demagnetization from 5 to 80 mT. A few specimens were demagnetized thermally. Both demagnetization methods gave the same results (Fig. 3). The magnetic behavior during demagnetization was straightforward. In low fields/temperatures (less than 20 mT or 200 °C), a viscous component which is often parallel to the present earth field (PEF) direction is removed in most sites ($N = 15$ sites, $D/I = 11°/57°$, $k = 34$, $\alpha_{95} = 6.6°$). Higher unblocking temperature or coercive force components decaying linearly to the origin are easily isolated from almost all the specimens. Unblocking temperatures near 575 °C and rapid saturation of isothermal remanent magnetizations of three specimens (saturation before 150 mT) from all three dike

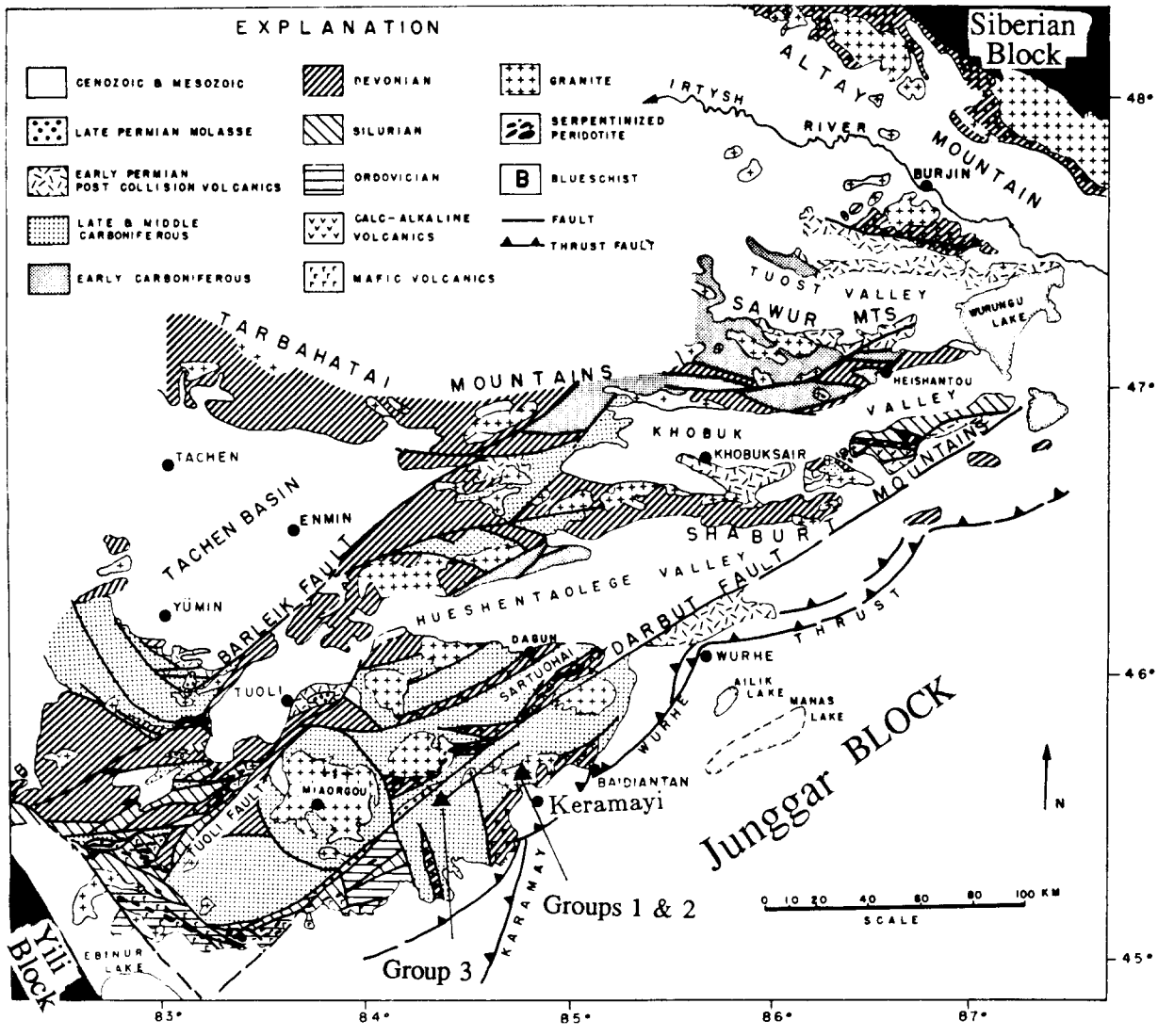


Fig. 2. Generalized geologic map of northwest Junggar Block (modified from Feng et al. [1]). Sampling localities are indicated by triangles.

swarms suggest that magnetite is the main magnetic carrier in these dikes. The α_{95} values for about 80% of the sites are less than 10°, indicating good internal consistency within each site. Sites 64 and 66 from the second dike swarm have large dispersions. Both polarities of magnetization are recorded by specimens from these two sites, producing the low k values. Because each dike should only be an instantaneous reading of the geomagnetic field, we believe that normal polarity components are simply the PEF overprint. These normal overprint components were not completely removed during the demagnetizations of these two sites, and therefore these sites were not included in the calculation of the final mean direction. Apart from these two sites, simple demagnetization behavior during the demagnetizations and reversed polarities (consistent with Late Paleozoic Kiaman Reversed Interval, 248–320 Ma) suggest that the isolated magnetizations are primary thermal remanent magnetizations formed when the dikes initially cooled.

At the 99% confidence level, no significant difference exists between the mean directions from

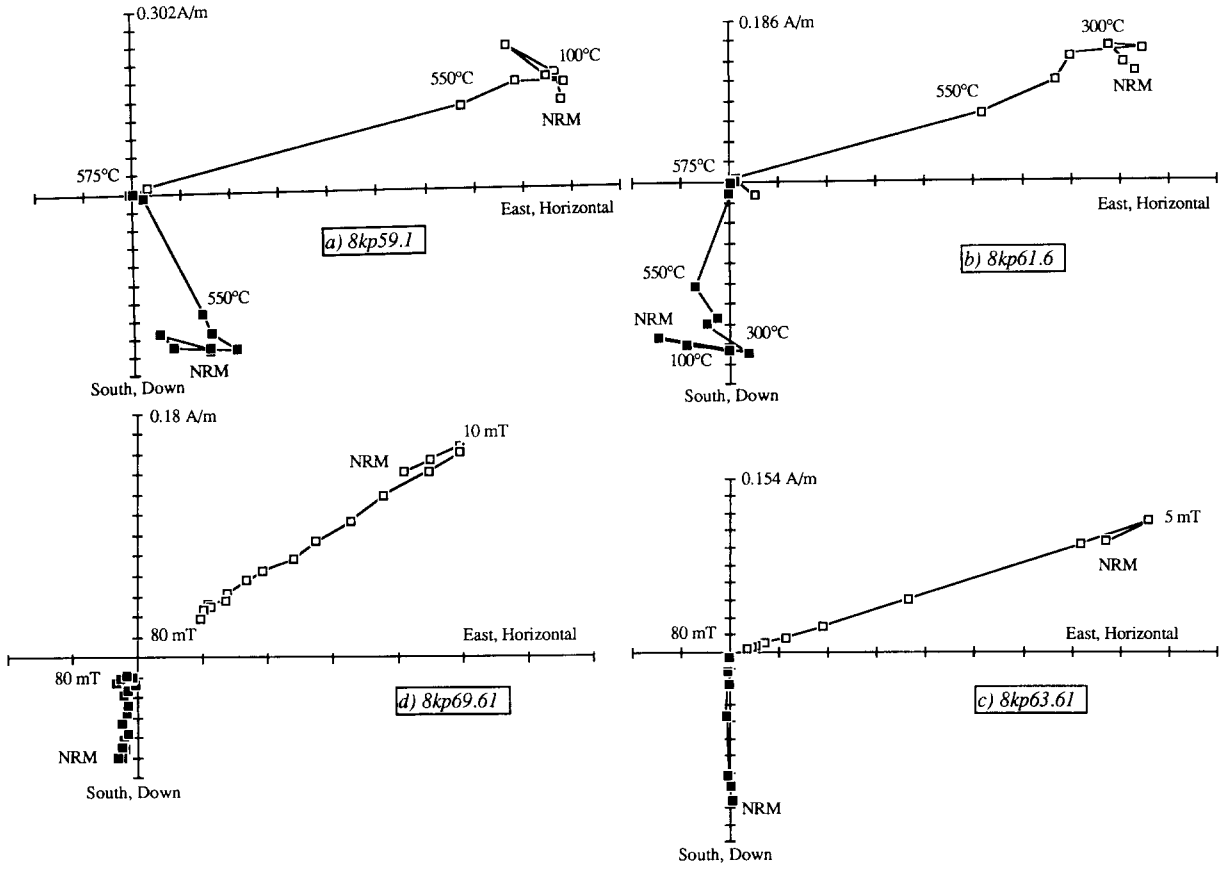


Fig. 3. Orthogonal vector plots of thermal and alternating field demagnetizations of the specimens from dikes near Keramayi, northwest Junggar Block. Filled symbols refer to the projection on the horizontal plane; open symbols refer to the projection on a vertical plane. Specimens 8kp59.1 and 8kp61.61 are from the first dike swarm; specimens 8kp63.61 and 8kp69.61 are from the second and third dike swarms respectively.

the first and second dike swarms [7] (Table 1). We believe that these two groups of directions belong to a single population, and thus these data were combined to yield one mean direction which is significantly different from that of the third dike swarm at the 99% confidence level. The difference is primarily in inclination, as the means of the two populations differ by 28° (Table 1 and Fig. 4). At present we do not know what causes the difference, but three explanations are possible:

(1) One possibility is that there is an age difference between the dike swarms, during which the Junggar Block had rapidly changed latitude. If so, the Junggar Block must have moved 20° in latitude during the Permian. The shift could be either north or south, depending upon the relative ages of the dike swarms. Subduction in this region had probably ended by the Permian, so it is unlikely that the Junggar Block moved rapidly after the closure of the ocean. Our unpublished Late Paleozoic paleomagnetic data from other parts of the Junggar Block (Hebukeshale, northwestern Junggar and Urumqi, southern Junggar) do not support rapid polar wander during the Late Paleozoic for the Junggar Block. We cannot rule out the possibility of a later remagnetization of the third dike swarm relative to the other two or vice versa, although regional geologic evidence shows no indication of thermal activity after the Permian in the sampling area and all stable magnetizations are reversed, consistent with a Kiaman age.

(2) Another possibility is geomagnetic secular variation. At a latitude of 35° (the mean paleolatitude derived from this data set), the virtual geomagnetic pole (VGP) angular dispersion S is

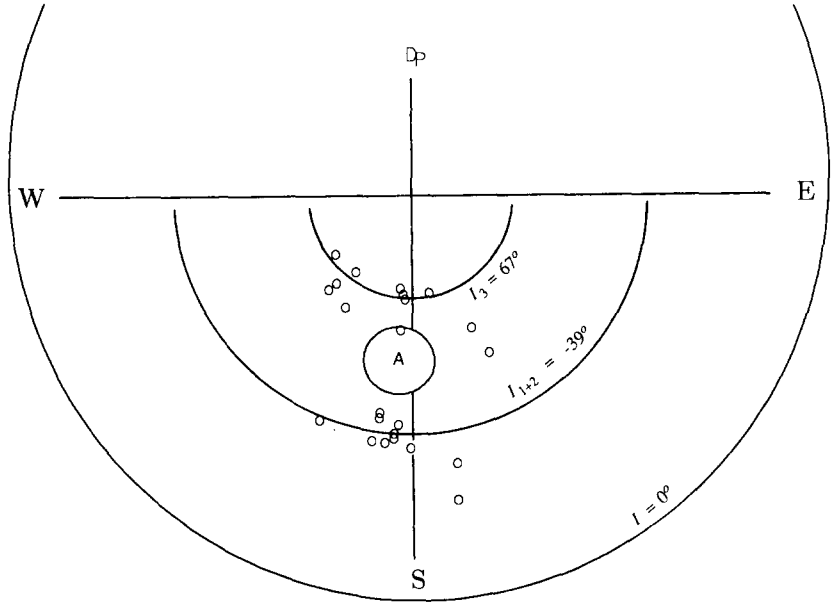


Fig. 4. Equal area stereographic projections of the site mean magnetic directions from Table 1. Open symbols refer to directions plotted on the upper hemisphere (negative inclination). *A* is the mean of all 23 sites for this study. *D* and *P* are the dipole field and present earth field directions at the sampling area plotted on the lower hemisphere. I_{1+2} and I_3 are the mean inclinations for the combination of the first and second dike swarms and the third dike swarm respectively.

about 16° based on model F [8], whereas *S* in our study, given by $81K^{-1/2}$, is 20°. It has been suggested that during times of low or zero reversal frequency (e.g. Cretaceous Normal and Kiaman Reversed Intervals), the non-dipole to dipole field ratio is reduced and in turn *S* may be smaller [9]. The directions from both the combined first and second dike swarms and the third dike swarm are not Fisherian; rather they vary more in declination that in inclination (Fig. 4). This kind of distribution could be caused by either strike-slip motions along the nearby faults or secular variation. The average within-site *k* value for each of the three dike swarms is 168, 162 and 494, whereas the comparable between-site *k* values are 112, 72 and 49, respectively. These values suggest that at least some secular variation is averaged within each dike swarm, and that all the dikes within each swarm were not emplaced simultaneously. But if secular variation is even partially averaged within each dike swarm, it seems less likely that the 28° inclination difference observed can result from secular variation alone.

(3) Finally, it is possible that the third dike swarm has been tilted relative to the other two dike swarms. To test this hypothesis, we plotted azimuths and dips of the dikes measured in the field (Figs. 5 and 6). The majority of the dike strikes for the third dike swarm cluster around N50° E (mode between N48° E and N52° E), while for the first and second dike swarms, the orientations of the dikes are more scattered. Regional tilt will result in the departure of the dike dips from vertical, except for tilt occurring about an axis perpendicular to the strike of the dikes, provided that the dikes intruded the plutons vertically. Most dips of the dikes are in excess of 85° in this study (Fig. 6). The pluton intruded by the first and second dike swarms probably has been tilted very little after the intrusion of the dike swarms because (a) the orientations of dikes are varied (Fig. 5), and (b) the Mesozoic lacustrine sequences around the pluton are only slightly tilted (< 15°) outward, as a result of uplift of the pluton. The pluton intruded by the third dike swarm is small, and could have tilted toward the northeast while the dips of the dikes have remained near vertical (Figs. 2 and 6). However, the amount of the tilting cannot be exactly inferred at present. To sum up, we believe that tilt and geo-

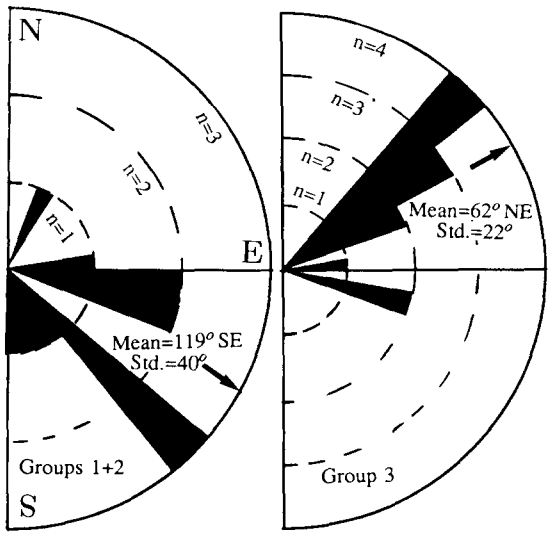


Fig. 5. Azimuthal distribution of dike strike.

magnetic secular variation are the causes of the inclination dispersion we observe.

4. Tectonic interpretation

If we assume that only secular variation causes the difference in inclination, the Late Paleozoic paleolatitude of the Junggar Block (at the sampling sites) was about 35° N ($I_{mean} = -54°$). Implicit in this assumption is the second assumption that geomagnetic secular variation may contribute to overall dispersion, but does not produce an inclination bias. The expected magnetic field directions from Late Carboniferous and Early Permian poles of Tarim and Kazakhstan are $D/I =$ 218°/−57° and 229°/−58° (reversed polarities) [10–13], suggesting an apparent consistency between the observed and expected inclinations. The tectonic implication from this comparison is that there has been no significant north-south motion of the Junggar Block relative to Tarim and Kazakhstan since the magnetization of the dikes. Current tectonic models based on geologic evidence suggest that the Junggar Block had accreted to Asia by the Permian [1–5]. These lines of geologic evidence include (1) Late Carboniferous post-collision granitic plutons which stitch the Kazakhstan and Junggar Blocks; (2) a regressive sequence from Carboniferous turbidites to Lower Permian shallow marine sandstones and Upper Permian fluvial and lacustrine deposits in southern Junggar; (3) substantial tectonic subsidence during the Late Permian indicated by thickness of sedimentary sequences in southern Junggar [5], and (4) Angara flora found throughout in the Junggar Block during the Permian [14].

Alternatively, if the third dike swarm has been tilted, the mean inclination of the first and second dike swarms ($I_{1+2} = -39°$) is shallower than the directions predicted from the Tarim and Kazakhstan Blocks, suggesting more than 1500 km northward motion of the Junggar Block with respect to the Tarim and Kazakhstan Blocks since the dikes were magnetized. At present, we cannot accurately determine the relative contributions to the inclination dispersion from either geomagnetic secular variation or local tilt.

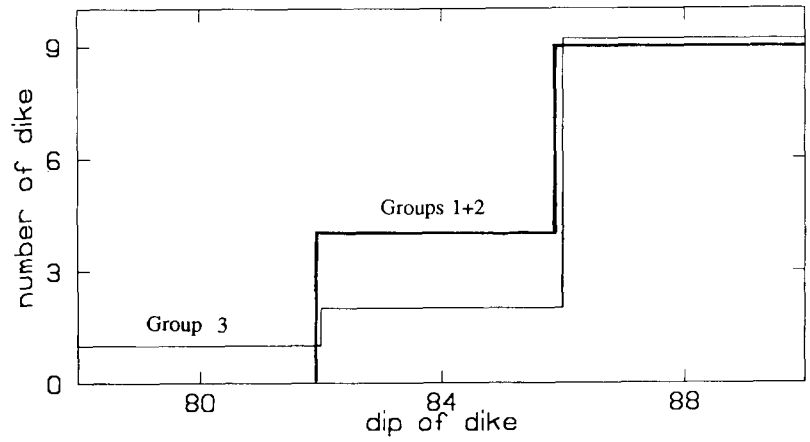


Fig. 6. Histogram of dike dip.

TABLE 1

Paleomagnetic data

Site	n/m	D (°)	I (°)	k	α_{95} (°)
First dike swarm (lat. = 45.7° N, long. = 82.7° E)					
55	7/7	183	−43	179	4.5
56	6/6	188	−45	254	4.2
57	6/6	189	−39	100	6.7
58	6/6	202	−40	179	5
59	6/5	170	−34	57	10
60	6/6	188	−44	101	6.7
61	6/6	184	−41	142	5.6
62	6/6	184	−40	332	3.7
$Mean_1$	8/8 *	186	−41	112	5.2
Second dike swarm (lat. = 45.7° N, long. = 82.7° E)					
63	6/6	180	−38	255	4.2
64′	6/6	19	75	19	15.8
65	6/6	171	−26	113	6.3
66′	4/6	41	69	119	18.0
67	4/6	186	−39	30	17.0
$Mean_2$	3/5 *	179	−35	72	14.7
Third dike swarm (lat. = 45.5° N, long. = 84.3° E)					
68	6/6	213	−73	2521	1.3
69	6/6	184	−69	434	3.2
70	5/6	170	−69	206	5.3
71	6/6	186	−70	773	2.4
72	6/6	228	−70	42	10.5
73	6/6	213	−70	27	13
74	5/5	183	−68	510	3.4
75	6/6	209	−63	99	6.8
76	6/6	218	−66	570	2.8
77	6/6	184	−62	163	5.3
78	5/6	154	−54	110	7.3
79	6/6	156	−60	468	3.1
$Mean_3$	12/12 *	190	−67	49	6.3
$Mean_{1+2}$	11/13 *	184	−39	87	4.9
$Mean_t$	23/25 *	186	−54	21	6.7

n/m: number of samples or sites (*) accepted/measured. Each site is an individual dike. Sites 64 and 66 are excluded in calculation (see text). k is Fisher's precision parameter; α_{95} is radius of the 95% confidence limit. $Mean_1$, $Mean_2$, $Mean_{1+2}$, $Mean_3$ and $Mean_t$ are the mean directions of the first, second, combination of first and second, third dike and overall dike swarms respectively.

Our paleomagnetic data do suggest that the sampling region has rotated 30–40° counterclockwise relative to Tarim or Kazakhstan as the declination values from all dike swarms are in agreement ($D_{mean} = 186°$). Although the regional extent and timing of the rotation is not known, the rotation is probably related to continued post-collision transcurrent and thrust faulting. Several faults near the sampling region (Barleik, Tuoli and Darbut, Fig. 2) are all high-angle, strike-slip faults which could have accomodated significant amounts of left-lateral displacement. Block rotations are associated with strike-slip motion [15], and the rotation observed could be related to this sinistral shear. Limited paleomagnetic data from granitic plutons bounded by these faults are discordant in declination (Li, Y-A., unpublished data). Seismic reflection sections and geologic mapping show low angle detachment faults thrusting toward the basin (southeast) in the Keramayi region [16]. Thrusting involving rocks as young as Triassic has produced horizontal displacements of 20 km and vertical displacements of less than 5 km [16], and rotation could have occurred during this thrusting process. Present northeast-trending ultrabasic bodies in western Junggar would be lined up with northwest-trending ultrabasic zones in Semipalatinsk (lat. 50° N, long. 80° E, U.S.S.R.) and eastern Junggar (lat. 45° N, long. 90° E) if these bodies in western Junggar have been counterclockwise rotated 50° (Fig. 7) [17]. Large clockwise and counterclockwise relative rotations are commonly found in the other parts of the Tianshan Ranges [18–20].

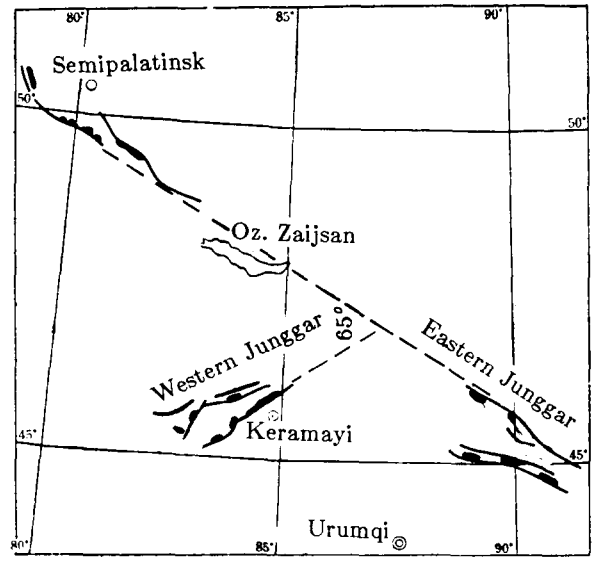


Fig. 7. Sketch map of distribution of ultramafic rocks in parts of central Asia (modified from Jiang [17]). Black areas are the outcrops of the ultramafic rocks. Thick lines are traces of the faults.

Acknowledgments

This work is a part of a collaborative paleomagnetic research project between the Department of Geophysics at Stanford University and Xinjiang National 305 Project Office of China. We thank R. Coleman for a preprint on the geology of the West Junggar region, and three anonymous reviewers. We thank Xinjiang Geology Institute (Urumqi) for help with sample collection. Financial support was provided in part by the Stanford-China Geosciences Industrial Affiliates Program participated by Amoco, Anschutz, Cheveron, ELF Aquitaine, Exxon, Mobil, Occidental, Pecten, Texaco, Unocal, School of Earth Sciences, Stanford University, National Science Foundation (EAR-8618562 and EAR-8718562) and the Xinjiang 305 Project Office.

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